

Campus Lost & Found App

Group 11

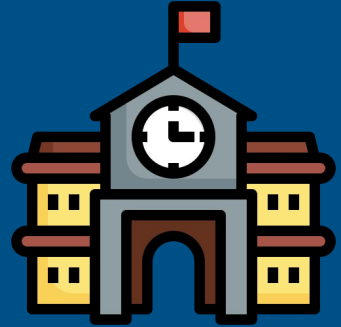
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Overview

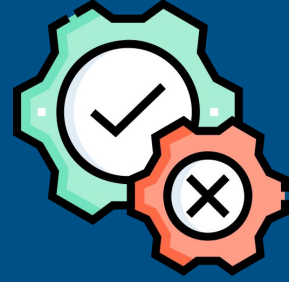
Overview - We will be creating a centralized mobile application for reporting, searching, and claiming lost items around campus.

Domain - College campuses are large, busy environments where personal items are frequently lost. Current solutions rely on social media posts or word of mouth, which are disorganized and inefficient.

Why this app is needed - Lost items cause stress, wasted time, and unnecessary replacement costs. Currently, there is no structured, searchable system that connects people who lose items with those who find them. So, our app provides a quick, reliable way for students, faculty, and staff to recover lost belongings.



Functional Requirements



DJ's Functional Tasks

- ❖ User registration and login/logout
- ❖ Profile creation
- ❖ Secure authentication and session handling
- ❖ “My Items” dashboard to manage user posts

Ally's Functional Tasks

- ❖ Lost item reporting workflow
- ❖ Update lost item status (active, found)
- ❖ Track claim requests submitted for a lost item
- ❖ Category tagging for lost items (electronics, clothing, etc.)

Malavika's Functional Tasks

- ❖ Found item reporting workflow
- ❖ Update found item status (available, claimed)
- ❖ Accept or decline claim requests
- ❖ Attach proof-of-ownership verification step before marking as claimed

Taeoh's Functional Tasks

- ❖ Admin dashboard
- ❖ Content moderation tools
- ❖ Inappropriate behavior reporting
- ❖ Analytics and system monitoring

Unnati's Functional Tasks

- ❖ Advanced search and filtering
- ❖ Item matching recommendations
- ❖ Notifications for matched items
- ❖ In-app messaging between users

Non-Functional Requirements

- ❖ **DJ:** Secure and encrypted data transmission with protected session management
- ❖ **Ally:** Reliable real-time data consistency for lost item updates
- ❖ **Malavika:** Fast and responsive UI for found item workflows
- ❖ **Unnati:** Scalable search and notification performance
- ❖ **Taeoh:** System reliability and administrative access control



Responsibilities

DJ – Frontend + Authentication Backend

Ally – Frontend + Backend for the Lost Items

Malavika – Frontend + Backend for the Found Items

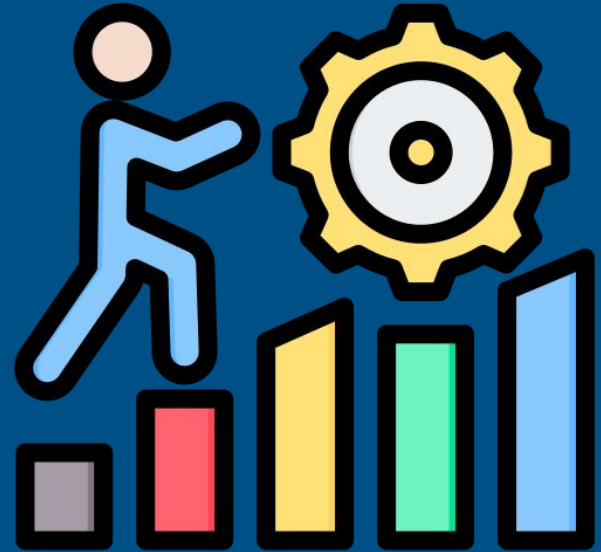
Unnati – Backend for Search, Matching & Communication

Taeoh – Front end for Admin and Backend for System Management



Potential Challenges

- **Real-Time Data Synchronization**
 - If two people find the same item or if an item is claimed but the post remains active, it leads to confusion. Managing concurrency—where multiple users are interacting with the same data entry simultaneously—can be tricky for a new dev team.
- **Image Recognition & Auto-Tagging**
 - Most people won't want to type a long description. You may face a steep learning curve in integrating Computer Vision APIs (like Google Vision or AWS Rekognition) to automatically tag an item (e.g., identifying a "Hydroflask" or "iPhone") just from a photo.



Strategies

➤ Real-Time Data Synchronization

- Include a status to indicate if a certain item is available, pending, or claimed.
- When a user starts the claim process it will be temporarily locked as pending.
- If a claim isn't completed within a set time window, the item automatically returns to available status, reducing abandoned claims.

➤ Image Recognition & Auto-Tagging

- Image recognition can be used to suggest tags like “water bottle,” “phone” which will allow users to confirm or edit. If image recognition fails or confidence is low, the app goes back to simple manual tagging so the functionality is never blocked.

